

SYSC 4504: Fundamentals of Web Development

Programming Assignment 2: Advanced JavaScript

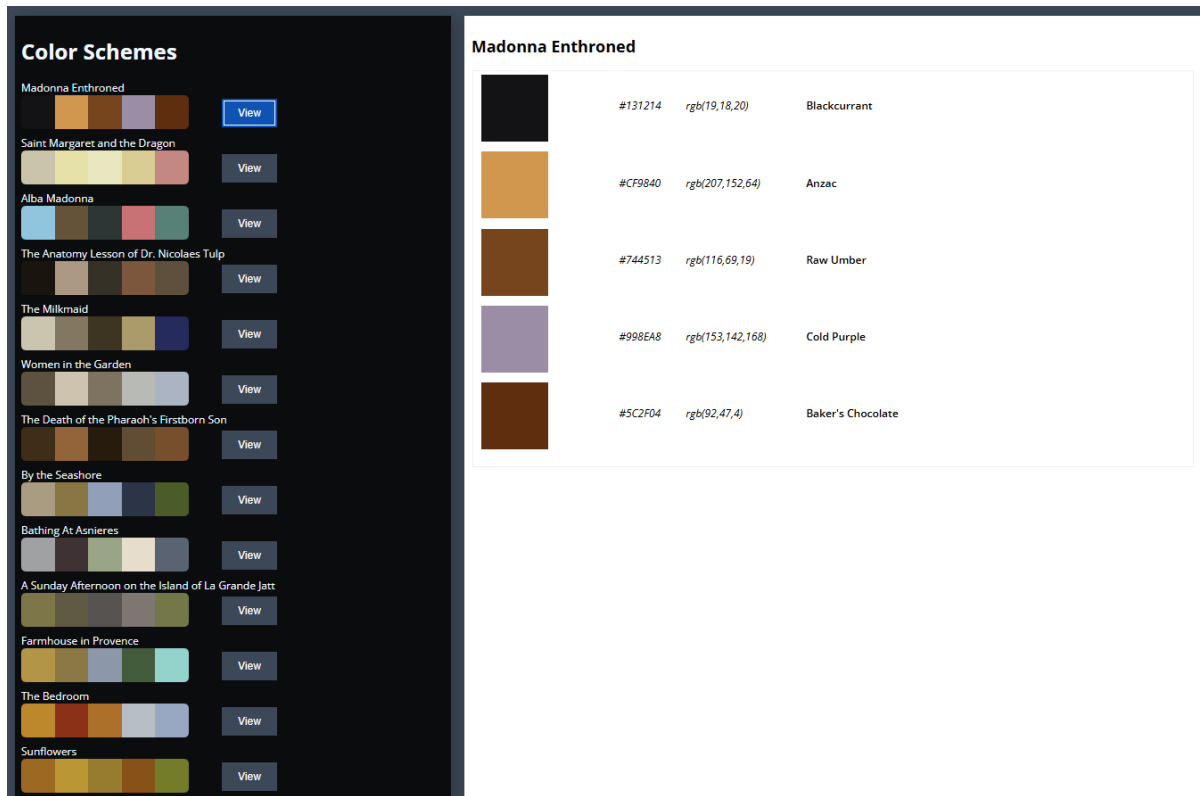
Due: Friday, February 27, 1 pm

Assignment Description

This project uses fetch to retrieve data and applies the DOM techniques discussed in class. An initial set of files is provided as ZIP archive on Brightspace, containing three files: `assignment2.html`, `assignment2.css`, and `assignment2.js`. By now you should know what sort of content each file contains. When you open `assignment2.html` (which loads the other two files) in a web browser, it should look as follows (here using Microsoft Edge):



Once you completed the assignment, loading `assignment2.html` and clicking on the “View” button next to Madonna Enthroned, the page will look as follows:



- You have been provided with the necessary styling and markup already. Examine `assignment2.html` in the editor of your choice. Notice the sample markup for the color scheme list items. This will be eventually commented out and replaced with JavaScript code that programmatically generates this markup. There is also a loading animation that will need to be displayed/hidden.
- Examine `assignment2.js` in the editor of your choice. In it, you will see the URL for the external API that will provide the color scheme data. Examine this URL in the browser in order to see the structure of the data. **The first time you access this data, your browser may complain about a certificate error. Proceed to the resource anyways, once you convinced your browser to accept the invalid/out-of-date certificate, you are good to go.**
- Fetch this scheme data from the API and display it within the `<article>` element. As you can see from the sample supplied markup, this will require creating `<h3>`, `<section>`, `<div>`, and `<button>` elements.
- Display the loading animation before the fetch and then hide it after the data is retrieved.
- Set up a single click event handler for all the View buttons. This will require using event delegation. When the user clicks a view button, display the scheme details in the `<aside>` element. As you can see from the sample supplied markup, this will require creating `<div>` elements within the supplied `<fieldset>`. You will also have to change the `<h2>` content to the clicked scheme name. Hint: use the `find()` method to retrieve the correct scheme object from the `data-id` property of the clicked button. Also, remember to clear out the previous content of the `<fieldset>` by setting its `innerHTML` to `""`.

Guidance and Testing

1. Break this problem down into smaller steps. First verify the fetch works, perhaps with a simple `console.log` statement. Then write a function that generates the markup for a single color scheme in the `<article>` and test to make sure it works. This will require a loop, so try using `forEach()` instead of a `for` loop.
2. Then add in support for the loading animation.
3. Before generating the scheme details, add in the event handler using event delegation and verify (again using `console.log`) if you are able to retrieve the correct scheme object using `find()`.
4. Finally, write a function that generates the scheme details. This will require a loop, so try using `forEach()` instead of a `for` loop.

Submission Requirements

Submit your assignment using Brightspace, under Assignment 2 Submission for the course. Submit a single ZIP file containing all files making up the solution to the assignment (so the file should contain at least `assignment2.html`, `assignment2.js`, and `assignment2.css`, if you created additional files that are loaded by the main page, include these in the ZIP file as well). Marks will be based on:

- Completeness of your submission
- Correct solution to the problem
- Adhering to the submission requirements

Similar to assignment 1, the due date is based on the time of the Brightspace server and will be strictly enforced. If you are concerned about missing the deadline: multiple submissions are allowed. So you can always submit a (partial) solution early, and resubmit an improved solution later. This way, you will reduce the risk of running late, for whatever reason (slow computers/networks, unsynchronized clocks, failure of the Internet connection at home, etc.).